Python for Data Analysis

Learn to turn a messy table into a clear answer, using the tools data analysts use.

Who it is for. S.4 to S.6 students and school leavers. No coding, spreadsheet or statistics experience is needed.

Tools. Python, Jupyter notebooks, NumPy, pandas, matplotlib and seaborn. All are free and used in industry.

What your child will build

Each learner analyses a set of data and answers one question. They clean it, keep a log of every change, summarise it, draw labelled charts, and check their totals two different ways. They present it in a five-minute talk and a short written report.

Practice data is made up for the camp, so no real person's information is used.

The eight weeks

1	**Meet the data**	Open a notebook, read a table of shop sales, store values and read error messages.
2	**Python you need**	Decisions, lists, loops and functions, with checks that stop the program when something is wrong.
3	**Tables in pandas**	Select, filter, sort and group a table. Revenue by item and by branch, checked by hand.
4	**Clean and chart**	Finding and fixing duplicates and gaps, bar, line, histogram and scatter charts. Checkpoint one.
5	**Explore with NumPy and seaborn**	Spread, typical values, unusual values, and why two things moving together do not prove one causes the other.
6	**Messy data and two tables**	Clean a raw export, join two tables, and explain any difference in the totals. Checkpoint two.
7	**Show it**	An approved project plan and a notebook that reruns. Modern AI assistants are shown and critiqued.
8	**Hand over**	The report, the five-minute talk, and each learner demonstrates their work to parents and guardians.

The practice data comes from a made-up chain of three shops, Sunrise Shops. Final projects may use public real data once the licence has been read at its source.

What a class looks like

Each three-hour class has the same shape. A short recall, then a guess written on paper before anything is run, then the code is run and investigated, a 15-minute break away from the screen, a second block, and a make time where each learner builds their own version. Every class ends with a sentence the learner writes in their own words.

Nobody is expected to know any of this beforehand. Every new word is explained with a picture and an everyday example, and the instructor works with each learner at their own pace.

How progress is checked

Every mark points to something the learner made. Marks are private, and learners are never ranked in public.

Six weekly portfolios	**20%**
Checkpoint one, end of week 4	**15%**
Checkpoint two, end of week 6	**20%**
Final project	**30%**
Individual demonstration	**15%**

To earn the certificate
At least 20 of the 24 classes attended or made up.
Both checkpoints, the final project and the demonstration handed in.
An overall mark of at least 50 out of 100.

Using AI tools

In weeks 1 to 6 learners write and fix their own code, so that the foundations are theirs. In weeks 7 and 8 they see how modern AI coding assistants work, and learn to check what such a tool says before trusting it. The final demonstration is done without AI help, and the certificate shows what the learner can do themselves.

What we ask of families

- Your child brings their own laptop and charger, Windows or Mac. We check it works with a short setup session before day one.
- Learners attend on time and tell us ahead of any absence, so a class can be made up.
- Parents and guardians come to the final demonstration in week 8.
- Photographs and sharing a project publicly are optional, and need a separate written consent.

Fee	UGX 1,000,000 per learner. The same for a group of one to five.
Venue	The National ICT Innovation Hub, Nakawa.
Schedule	Three classes a week for eight weeks, starting December 2026. Dates and times are confirmed at enrolment.
Time	24 classes of 3 hours: 72 scheduled hours including breaks.
Groups	No more than five learners in a group, so each one gets personal attention.

Your tutor. Jotham Martin Wambi, data scientist and ML/AI engineer, AI and ML lecturer at IUEA.

This is a beginner programme. It builds real skills and a project to show, and is not a professional qualification.